

Musterlösung Serie 14

INVERSES, DUAL VECTOR SPACE

This exercise sheet is to be handed in the week before the Frühjahr Semester 2026.

We repeat here the first exercise from Sheet 13 as the techniques to solve it were only partially covered in the lecture.

1. Inverses. For each of the following matrices, determine whether or not it is invertible, and if it is compute its inverse.

(a) $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

(b) $\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$

(c) $\begin{pmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix}$

(d) $\begin{pmatrix} 1 & 2 & -3 & 1 \\ -1 & 3 & -3 & -2 \\ 2 & 0 & 1 & 5 \\ 3 & 1 & -2 & 6 \end{pmatrix}$

Lösung:

- (a) This matrix is of rank 1 since the first row is equal to the second and to the third. Therefore it cannot be transformed into the identity using elementary row-operations and it isn't invertible.

(b) $\begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$

$$(c) \begin{pmatrix} 4/5 & -1/5 & -1/5 & -1/5 \\ -1/5 & 4/5 & -1/5 & -1/5 \\ -1/5 & -1/5 & 4/5 & -1/5 \\ -1/5 & -1/5 & -1/5 & 4/5 \end{pmatrix}$$

$$(d) \begin{pmatrix} 35/11 & -16/11 & 13/11 & -2 \\ 1 & 0 & 1 & -1 \\ 10/11 & -3/11 & 10/11 & -1 \\ -16/11 & 7/11 & -5/11 & 1 \end{pmatrix}.$$

2. Dual and bidual. Let $(v_1, \dots, v_n)$ be an ordered basis of a vector space V , let $(v_1^*, \dots, v_n^*)$ the corresponding dual basis of the dual vector space V^* , and let $(v_1^{**}, \dots, v_n^{**})$ the dual basis corresponding to B^* of the bidual vector space $(V^*)^*$. Show, that the natural isomorphism

$$\tau : V \xrightarrow{\sim} (V^*)^*, \quad v \mapsto \tau(v)$$

maps every v_j to the corresponding v_j^{**} .

Lösung: The dual basis is characterised by the condition $v_i^*(v_j) = \delta_{i,j}$ for all i, j . Similarly, we have $v_j^{**}(v_i^*) = \delta_{i,j}$ for all i, j . Moreover, the evaluation map is defined by $\tau(v)(\ell) = \ell(v)$ for all $v \in V$ and $\ell \in V^*$. For every pair i, j , this yields

$$\tau(v_j)(v_i^*) = v_i^*(v_j) = \delta_{i,j} = v_j^{**}(v_i^*).$$

Hence the linear maps $\tau(v_j) : V^* \rightarrow K$ and $v_j^{**} : V^* \rightarrow K$ are equal on the bases $(v_1^*, \dots, v_n^*)$ of V^* and are thus equal.

3. Dual and Hom. Let V and W be finite-dimensional vector spaces over a field K . Show that:

$$(a) \operatorname{Hom}(V, W) \cong \operatorname{Hom}(W^*, V^*);$$

(b) The following map is an isomorphism

$$\begin{aligned} \operatorname{Hom}(V, V) &\rightarrow \operatorname{Hom}(V, V)^* \\ T &\mapsto [S \mapsto \operatorname{tr}(S \circ T)], \end{aligned}$$

where $\operatorname{tr} \in \operatorname{Hom}(V, V)^*$ is the *trace map* defined by

$$\operatorname{tr}(T) = \operatorname{tr}([T]_{\mathcal{B}}^{\mathcal{B}}) = \sum_{i=1}^n ([T]_{\mathcal{B}}^{\mathcal{B}})_{ii},$$

for any basis $\mathcal{B}$ of V .

Remark. You may use this in the above exercise, but as a bonus you could first show that the map $T \in \operatorname{Hom}(V, V) \mapsto \operatorname{tr}(T)$ is independent of the choice of basis of V .

Lösung:

(a) Consider the map

$$\begin{aligned} \Phi : \text{Hom}(V, W) &\rightarrow \text{Hom}(W^*, V^*) \\ T &\mapsto [\Phi(T) : \ell \mapsto \ell \circ T, \forall \ell \in W^*] \end{aligned}$$

We first check that Φ is well-defined. Let $T \in \text{Hom}(V, W)$, $\ell, \ell' \in W^*$, and $\alpha \in K$. Then,

$$\begin{aligned} \Phi(T)(\ell + \alpha\ell') &= (\ell + \alpha\ell') \circ T \\ &= \ell \circ T + \alpha(\ell' \circ T) \\ &= \Phi(T)(\ell) + \alpha\Phi(T)(\ell'). \end{aligned}$$

So, $\Phi(T)$ is indeed an element of $\text{Hom}(W^*, V^*)$ for all $T \in \text{Hom}(V, W)$.

Let us now show that Φ is injective. Assume that $\text{Im}(T) \neq \{0\}$ and let $w \in \text{Im}(T) \setminus \{0\}$. Extend $\{w\}$ to a basis $\mathcal{B}$ of W . Now define a map ℓ on $\mathcal{B}$ by setting

$$\ell(w) = 1 \quad \text{and} \quad \forall w' \in \mathcal{B} \setminus \{w\} : \ell(w') = 0$$

and extend it linearly to a functional of W . Then $\text{Im}(T) \not\subseteq \ker(\ell)$ implies that $\Phi(T)(\ell)$ does not vanish on the whole of V and therefore that $\Phi(T)$ is not the 0 map. This shows that Φ is injective.

Now, let $\eta \in \text{Hom}(W^*, V^*)$, let $\{e_i\}_{i=1}^{\dim(W)}$ be a basis of W , and define a map $T : V \rightarrow W$ by letting

$$T(v) := \sum_{i=1}^{\dim(W)} \eta(e_i^*)(v) e_i.$$

Since the sum is finite and since $\eta(e_i^*)$ is linear for all i , $T \in \text{Hom}(V, W)$. Additionally, for any $\ell \in W^*$, for any $v \in V$

$$\begin{aligned} \Phi(T)(\ell)(v) &= \ell \circ T(v) = \sum_{i=1}^{\dim(W)} \eta(e_i^*)(v) \ell(e_i) \\ &= \sum_{i=1}^{\dim(W)} \eta(\ell(e_i) e_i^*)(v) \\ &= \eta \left(\sum_{i=1}^{\dim(W)} \ell(e_i) e_i^* \right) (v) \\ &= \eta(\ell)(v), \end{aligned}$$

since $\sum_{i=1}^{\dim(W)} \ell(e_i) e_i^* = \ell$. This shows that Φ is surjective.

(b) Let us first show that the trace of an application is well-defined. We will use

Lemma 1. For any $A, B \in M_{n \times n}(K)$, we have

$$\operatorname{tr}(AB) = \operatorname{tr}(BA).$$

Beweis. This follows from a straight-forward computation. Denote by a_{ij} , respectively b_{ij} , the entries of A , respectively of B . Then the diagonal entries of the product AB are given by

$$(AB)_{ii} = \sum_{k=1}^n a_{ik} b_{ki}, \quad 1 \leq i \leq n.$$

So,

$$\begin{aligned} \operatorname{tr}(AB) &= \sum_{i=1}^n \sum_{k=1}^n a_{ik} b_{ki} \\ &= \sum_{i=1}^n \sum_{k=1}^n b_{ki} a_{ik} \\ &= \sum_{k=1}^n \sum_{i=1}^n b_{ki} a_{ik} = \operatorname{tr}(BA). \end{aligned}$$

□

The independence of the choice of basis now follows. Observe that if $\mathcal{B}, \mathcal{C}$ are arbitrary bases of V and $T \in \operatorname{Hom}(V, V)$ we have

$$\begin{aligned} \operatorname{tr}([T]_{\mathcal{C}}^{\mathcal{C}}) &= \operatorname{tr}([\operatorname{id}_V]_{\mathcal{C}}^{\mathcal{B}} [T]_{\mathcal{B}}^{\mathcal{B}} [\operatorname{id}_V]_{\mathcal{B}}^{\mathcal{C}}) \\ &= \operatorname{tr}([\operatorname{id}_V]_{\mathcal{C}}^{\mathcal{B}} [T]_{\mathcal{B}}^{\mathcal{B}} ([\operatorname{id}_V]_{\mathcal{C}}^{\mathcal{B}})^{-1}) \\ &= \operatorname{tr}([\operatorname{id}_V]_{\mathcal{C}}^{\mathcal{B}} ([\operatorname{id}_V]_{\mathcal{C}}^{\mathcal{B}})^{-1} [T]_{\mathcal{B}}^{\mathcal{B}}) \\ &= \operatorname{tr}([T]_{\mathcal{B}}^{\mathcal{B}}), \end{aligned}$$

where the second-to-last equality follows from ??.

We now show that the map

$$\begin{array}{ccc} \operatorname{Hom}(V, V) & \rightarrow & \operatorname{Hom}(V, V)^* \\ T & \mapsto & [S \mapsto \operatorname{tr}(S \circ T)], \end{array}$$

is well-defined. This means showing that the map

$$\begin{array}{ccc} \operatorname{tr}_T : \operatorname{Hom}(V, V) & \rightarrow & K \\ S & \mapsto & \operatorname{tr}(S \circ T) \end{array}$$

is linear. We will need the following result:

Lemma 2. *The map $\text{tr} : M_{n \times n}(K) \rightarrow K$ is linear. Namely, for any $A, B \in M_{n \times n}(K)$, and any $\alpha \in K$, the following hold:*

$$\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B) \quad \text{and} \quad \text{tr}(\alpha A) = \alpha \text{tr}(A).$$

Beweis. Denote $A = (a_{ij})$ and $B = (b_{ij}) \in M_{n \times n}(K)$. We have

$$\begin{aligned} \text{tr}(A + B) &= \sum_{k=1}^n (A + B)_{kk} \\ &= \sum_{k=1}^n a_{kk} + b_{kk} \\ &= \sum_{k=1}^n a_{kk} + \sum_{k=1}^n b_{kk} = \text{tr}(A) + \text{tr}(B). \end{aligned}$$

We now consider αA . We have,

$$\begin{aligned} \text{tr}(\alpha A) &= \sum_{k=1}^n (\alpha A)_{kk} \\ &= \sum_{k=1}^n \alpha a_{kk} \\ &= \alpha \sum_{k=1}^n a_{kk} = \alpha \text{tr}(A). \end{aligned}$$

□

Linearity of tr_T . Let $S, S' \in \text{Hom}(V, V)$ and $\alpha \in K$, then

$$\begin{aligned} \text{tr}_T(S + \alpha S') &= \text{tr}((S + \alpha S') \circ T) \\ &= \text{tr}(S \circ T + \alpha(S' \circ T)) \\ &= \text{tr}([S \circ T + \alpha(S' \circ T)]_{\mathcal{B}}^{\mathcal{B}}) \\ &= \text{tr}([S]_{\mathcal{B}}^{\mathcal{B}}[T]_{\mathcal{B}}^{\mathcal{B}} + \alpha[S']_{\mathcal{B}}^{\mathcal{B}}[T]_{\mathcal{B}}^{\mathcal{B}}) \\ &= \text{tr}([S]_{\mathcal{B}}^{\mathcal{B}}[T]_{\mathcal{B}}^{\mathcal{B}}) + \alpha \text{tr}([S']_{\mathcal{B}}^{\mathcal{B}}[T]_{\mathcal{B}}^{\mathcal{B}}) \\ &= \text{tr}_T(S) + \alpha \text{tr}_T(S'). \end{aligned}$$

We deduce that $\text{tr}_T \in \text{Hom}(V, V)^*$.

Injectivity. We now show that the above map is injective. Fix T and assume that tr_T is the 0 map. Then for all $S \in \text{Hom}(V, V)$,

$$0 = \text{tr}([S]_{\mathcal{B}}^{\mathcal{B}}[T]_{\mathcal{B}}^{\mathcal{B}}) = \sum_{i=1}^n \sum_{k=1}^n s_{ik} t_{ki},$$

where we used the notation $T = (t_{ij})$ and $S = (s_{ij})$. Fix $(m, n) \in \{1, \dots, n\}^2$. Define $S_{m,n} = (\tilde{s}_{ij}) \in \text{Hom}(V, V)$ such that all the entries of $[S_{m,n}]_{\mathcal{B}}^{\mathcal{B}}$ vanish except for $\tilde{s}_{mn} = 1$. We deduce that

$$0 = \text{tr}([S_{m,n}]_{\mathcal{B}}^{\mathcal{B}}[T]_{\mathcal{B}}^{\mathcal{B}}) = t_{nm}.$$

Since (m, n) was arbitrary, this shows that $[T]_{\mathcal{B}}^{\mathcal{B}} = 0$ and therefore that T is the 0 map.

To conclude, remember that V is finite -dimensional, which implies that $\text{Hom}(V, V)$ is too. It follows that $\dim(\text{Hom}(V, V)) = \dim(\text{Hom}(V, V)^*)$ and hence, since the given map is injective, it is an isomorphism.

Single Choice. In each exercise, exactly one answer is correct.

1. Let $\dim V = 4$. Then there exists $\varphi \in V^*$ with $\dim \operatorname{Ker} \varphi = 2$.

(a) Correct

✓ False

Solution: $\varphi \in V^*$ means that $\varphi \in \operatorname{Hom}_K(V, K)$. If now $\dim \operatorname{Ker} \varphi = 2$ was true, the dimension formula

$$\dim V = \dim \operatorname{Im} \varphi + \dim \operatorname{Ker} \varphi$$

would yield $\dim \operatorname{Im} \varphi = 2$. This contradicts $\dim \operatorname{Im} \varphi \leq \dim K = 1$.

2. Every finite dimensional vector space is the dual of another finite dimensional vector space.

✓ Correct

(a) False

3. The set of all invertible $n \times n$ -matrices is...

✓ not a real linear subspace of $M_n(\mathbb{R})$

(a) a real linear subspace of $M_n(\mathbb{R})$

Solution: The set is not closed under addition. Indeed, we have

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + (-1) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

and the zero matrix is not invertible.

4. Let $f : V \rightarrow W$ be an arbitrary homomorphism between two K -vector spaces. Which of the following five assertions is not equivalent to the others?

(a) f is injective.

(b) The dual map $f^* : W^* \rightarrow V^*$ is surjective.

(c) The zero element of V is the only element mapped to the zero element of W .

✓ There exists a Homomorphism $g : W \rightarrow V$ with $f \circ g = \operatorname{id}_W$.

(d) For every $v \in V \setminus \{0\}$ there exists $\ell \in W^*$ with $\ell(f(v)) \neq 0$.

(e) All five assertions are equivalent.

Solution: Assertion (d) is equivalent to f being surjective, but not equivalent to injectivity and hence not to (a). Assertion (a) is equivalent to $\text{Kern}(f) = 0$ and thus to (c). As the exercise is correctly stated, (d) must be the correct answer. Indeed, (a) being equivalent to (b) was proved in exercise 17 of the repetition exercise sheet. Moreover, an element $w \in W$ is not equal to zero if and only if there exists $\ell \in W^*$ with $\ell(w) \neq 0$. Hence (e) is equivalent to $\forall v \in V \setminus \{0\} : f(v) \neq 0$, and hence equivalent to (a).

Multiple Choice.

1. For what value of parameter x is matrix $A = \begin{pmatrix} 1 & x & 1 \\ 3 & 3 & x \\ 0 & 3 & 1 \end{pmatrix}$ not invertible?

(a) 0

(b) 1

✓ 2

(c) 3

(d) 4